LIGHTSHIP WEIGHT ESTIMATION BOOKLET

Vessel Name:	M. V. YOUNUS MOLLA-1	Operating Route:	Inland & bay crossing
Vessel Type:	Steel Body Cargo Vessel	Plying Time:	Day & Night
Hull Construction:	Single Hull	Statutory Factor kt / kz:	1.40 / 1.40
Length Over All (LOA):	76.21 m	Breadth Moulded (B):	15.24 m
Length BP (LBP):	69.85 m	Depth Moulded (D):	4.87 m
Design Draft (d):	4.00 m	Block Coeff (Cb):	0.721
Transverse Frame Spacing:	550 mm	Structural Steel Density:	7,850 kg/m³

NAVAL ARCHITECTURE DESIGNER	INSPECTION & APPROVAL AUTHORITY
ROUNAK SAHA NILOY Naval Architect and Marine Engineer Designer ShipDyn Ltd.	Amitavo Chowdhury Wye Authorized Person ShipDyn Ltd.

1. STEEL PLATING WEIGHT ESTIMATION

Scantlings Plating Volume and Weight Evaluation | Mild Steel Density: 7,850 kg/m³

S/N	ELEMENT (Clean Plate Name)	L (mm)	B (mm)	t (mm)	Vol (m³)	Density	Qty	Weight (kg)	Weight (t)
1	Keel Plate	76,210	600	12.0	0.5487	7850	2	8,615	**8.615**
2	Bottom Shell Plating	76,210	5,200	12.0	4.7555	7850	2	74,661	**74.661**
3	Bilge Plating	76,210	3,000	14.0	3.2008	7850	2	50,253	**50.253**
4	Side Shell Plating	76,210	2,870	12.0	2.6247	7850	2	41,207	**41.207**
5	Sheer Strake Plating	76,210	900	12.0	0.8231	7850	0	6,461	**6.461**
6	Bulwark Plating	47,740	900	6.0	0.2578	7850	2	4,047	**4.047**
7	Hatchway Deck Plating	17,350	1,950	13.0	0.4398	7850	2	6,905	**6.905**
8	Main Deck Plating	15,753	15,240	9.0	2.1607	7850	1	16,961	**16.961**
9	Main Deck Wall	8,803	2,880	12.0	0.3042	7850	2	4,776	**4.776**
10	Poop Deck Wall	15,499	3,000	12.0	0.5580	7850	2	8,760	**8.760**
11	Poop Deck Plating	15,499	13,410	9.0	1.8706	7850	2	29,368	**29.368**
12	Bridge Deck Wall	7,840	2,500	12.0	0.2352	7850	1	1,846	**1.846**
13	Bridge Deck Plating	7,840	7,840	9.0	0.5532	7850	2	8,685	**8.685**
14	Transverse Watertight Bulkheads	15,240	4,870	10.0	0.7422	7850	5	29,131	**29.131**
15	Cargo Hatch Covers Plating	10,111	5,670	13.0	0.7453	7850	6	35,103	**35.103**
16	Forecastle Deck Plating	10,111	12,000	9.0	1.0920	7850	1	8,572	**8.572**
17	Double Bottom Tank Top Plate	76,210	15,240	12.0	13.9373	7850	0	109,408	**109.408**
18	Stem, Stern & Rudder Assembly	8,799	2,100	16.0	0.2956	7850	1	2,321	**2.321**
19	Superstructure Partition Bulkheads	15,499	2,550	12.0	0.4743	7850	2	7,446	**7.446**
	TOTAL STEEL PLATING WEIGHT:							**454,528 kg**	**454.528 t**

2. STEEL STIFFENERS & STRUCTURAL ANGLES WEIGHT ESTIMATION

Stiffener Scantlings Geometry & Unsupported Span Evaluation (Length linked to attached plate)

S/N	ELEMENT (Structural Profile)	L (mm)	B (mm)	t (mm)	Area (mm²)	Length	Vol (m³)	Density	Qty	Weight (kg)	Weight (t)
1	Center Keelson	400	250	12.0	7,800	76,210	0.5944	7850	1	4,666	**4.666**
2	Side Keelson	400	100	11.0	5,500	76,210	0.4192	7850	8	26,323	**26.323**
3	Side Stringer	250	100	10.0	3,500	76,210	0.2667	7850	6	12,563	**12.563**
4	Deck Girder	300	175	11.0	5,225	7,840	0.0410	7850	6	1,929	**1.929**
5	Floor	400	180	11.0	6,380	7,620	0.0486	7850	105	40,071	**40.071**
6	Web Frame	250	100	11.0	3,850	2,870	0.0111	7850	35	3,036	**3.036**
7	Main Frame	200	100	11.0	3,300	2,870	0.0095	7850	105	7,806	**7.806**
8	Deck Beam	125	75	9.0	1,800	1,250	0.0022	7850	52	918	**0.918**
9	Deck Web	300	175	11.0	5,225	7,840	0.0410	7850	13	4,180	**4.180**
10	Deck Beam (Main Deck)	75	75	6.0	900	1,250	0.0011	7850	30	265	**0.265**
11	Deck Beam (Poop Deck)	65	65	6.0	780	1,250	0.0010	7850	15	115	**0.115**
12	Deck Beam (Bridge Deck)	50	50	5.0	500	1,250	0.0006	7850	7	34	**0.034**
13	BHD Angle (Vertical)	120	120	10.0	2,400	4,870	0.0117	7850	45	4,129	**4.129**
14	BHD Angle (Horizontal)	120	120	10.0	2,400	15,240	0.0366	7850	45	12,920	**12.920**
15	Hatch Girder	65	65	6.0	780	17,350	0.0135	7850	15	1,594	**1.594**
16	Superstructure Girder (Main Deck)	75	75	6.0	900	7,840	0.0071	7850	4	222	**0.222**
17	Superstructure Girder (Poop Deck)	65	65	6.0	780	7,840	0.0061	7850	4	192	**0.192**
18	Superstructure Girder (Bridge Deck)	50	50	5.0	500	7,840	0.0039	7850	4	123	**0.123**
19	Superstructure Long. Girder (Main Deck)	200	90	7.0	2,030	7,840	0.0159	7850	1	125	**0.125**
20	Superstructure Long. Girder (Poop Deck)	200	90	7.0	2,030	7,840	0.0159	7850	1	125	**0.125**
21	Superstructure Long. Girder (Bridge Deck)	200	90	7.0	2,030	7,840	0.0159	7850	1	125	**0.125**
	TOTAL STEEL STIFFENERS WEIGHT:									**121,463 kg**	**121.463 t**

3. STATUTORY LIGHTSHIP SYNTHESIS & TOTAL WEIGHT ESTIMATION

Weight Item / Statutory Allowance Category	Basis Formula	Weight (Tonne)
Total Net Structural Steel (Plates + Angles)	W_plates + W_angles	**575.990 t**
Hull Allowance / Rolling & Welding Margin (20%)	Net Steel × 20%	115.198 t
Outfit & Deck Equipment (15%)	Net Steel × 15%	86.399 t
Piping, Valves & Fittings (10%)	Net Steel × 10%	57.599 t
Electrical & Instrumentation (3% formula / 5% rule base)	Net Steel × 3%	17.280 t
Accommodation Fittings & Furniture (1%)	Net Steel × 1%	5.760 t
Miscellaneous Statutory Items (7%)	Net Steel × 7%	40.319 t
TOTAL LIGHTSHIP WEIGHT (LIGHTWEIGHT DISPLACEMENT) :	**Net Steel × 1.56 (56% Outfitting)**	**898.545 Tonne**

